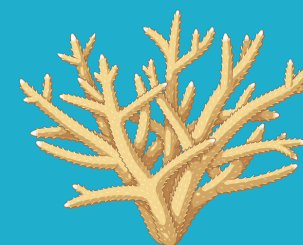




How do tidal and environmental conditions influence phosphate concentrations in reef habitats over time?



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Data

Experimental and in situ seawater nutrient data collected at Heron Island Research Station, southern Great Barrier Reef in 2021

Variables of interest include:

- PO_4 concentration (ppm)
- Tidal height (m)
- Time (days)

Methods

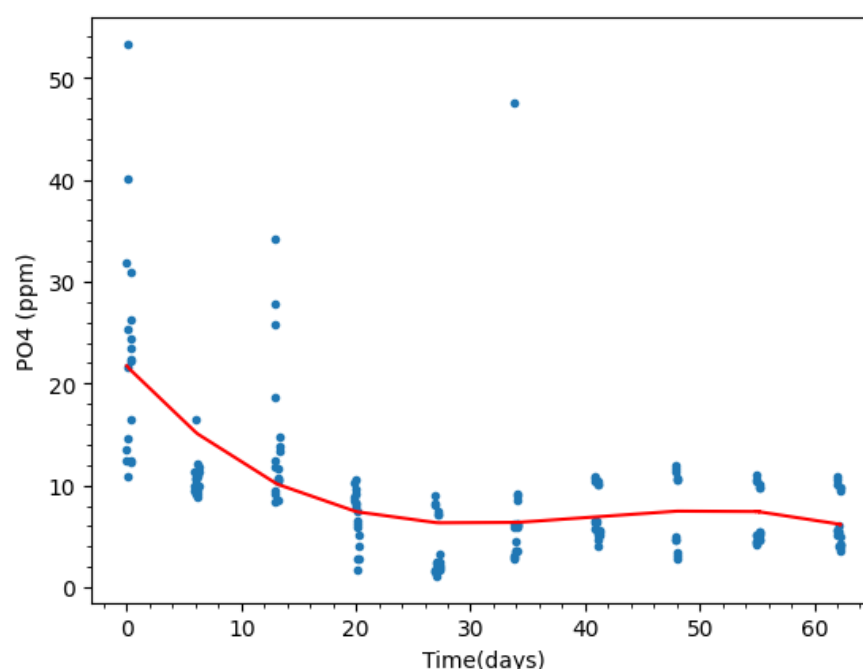
- Time series analysis
- Bayesian regression modelling

Relevance

- Phosphate is a biolimiting nutrient
- Forms the basis of the global food web, drives the oceans biological pump & influences global carbon cycling

Results

- Long term trend best fitted by a cubic model.
- PO_4 concentrations decrease from January to March.



- Phosphate concentration is higher at low tide.

